Temple of Doom – Machine Learning (MDS + BDMA + MIRI)

Duration: 2h

This is the most difficult theoretical and hand-computation exam. Full derivations required. No explanations without math.

Question 1

Let $D = \{(x_i, y_i)\}_{i=1}^{3} = \{(1, 1), (2, 2), (3, 1)\}$. Fit a linear model $f(x) = wx + b$ by least squares.

(a) Derive the closed-form solution for w and b.

(b) Compute the leave-one-out cross-validation (LOOCV) mean squared error by hand.

(c) Compare this with the training error. What do you observe?

Question 2

Let $f(x) = \sigma(w^T x + b)$ be a logistic regression model, where $\sigma(z) = \frac{1}{1+e^{-z}}$.

(a) Derive the gradient of the negative log-likelihood loss for a single sample (x, y).

(b) Show that this gradient is always bounded in magnitude.

(c) Compute the gradient explicitly for $x = [1, 2]^T$, $y = 1$, $w = [0, 0]^T$, $b = 0$.

Question 3

Let $k(x, x') = (x^T x' + 1)^2$ be a second-degree polynomial kernel. Define the implicit feature mapping $\phi(x)$ such that $k(x, x') = \phi(x)^T \phi(x')$.

(a) Explicitly write $\phi(x)$ when $x \in \mathbb{R}^2$.

(b) Verify your mapping by computing $k(x, x')$ for $x = [1, 2]^T$ and $x' = [3, 4]^T$ using both sides.

Question 4

Let $L(w) = \frac{1}{2n} \sum_{i=1}^{n} (y_i - w^T x_i)^2 + \frac{\lambda}{2} \|w\|^2$ be the ridge regression objective.

(a) Derive the closed-form solution for w.

(b) Under what conditions is this solution guaranteed to exist and be unique?

(c) Explain what happens to the solution when $\lambda \to 0$ and $\lambda \to \infty$.

Question 5

Manual Backpropagation: You are given a 1-hidden-layer MLP:

$$\text{Input: } \boldsymbol{x} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \boldsymbol{W}_1 = \begin{bmatrix} 2 & -3 \end{bmatrix}, \quad b_1 = 0, \quad \sigma(z) = \frac{1}{1+e^{-z}}, \quad W_2 = 1, \quad b_2 = 0$$

Loss: binary cross-entropy. Ground truth $y = 1$.

(a) Compute the forward pass and output $\hat{y}$.

(b) Compute $\frac{\partial L}{\partial W_1}$ and $\frac{\partial L}{\partial W_2}$ by hand using backpropagation.

Question 6

Consider a k-NN model with $k = 3$. Given the following points:

$$\text{Class 0: } (0,0), (0,1), (1,0) \quad \text{Class 1: } (3,3), (3,4), (4,3)$$

Predict the class of point $x = (2,2)$ with and without feature scaling. Justify numerically.

Question 7

You are told that a model performs well on the training set but very poorly on a test set drawn from the same distribution.

(a) List all possible causes assuming no bugs or leakage.

(b) Provide theoretical justifications for why each would create the observed outcome.

Question 8

Given a decision tree of depth 2, how many linear regions can it split $\mathbb{R}^2$ into at most?

(a) Justify with an example and a drawing.

(b) Compare this to the expressivity of logistic regression.

Question 9

Prove that the training error of 1-NN is always zero (assuming no ties), and the leave-one-out error is equal to the test error on the training set with each point left out.

Then:

(a) Explain why this makes LOOCV a pessimistic estimate for 1-NN.

Question 10

Let X be centered data. Show that in linear regression with no intercept term, the solution $\hat{w}$ satisfies the normal equation $X^T X \hat{w} = X^T y$.

Then:

(a) Solve for $\hat{w}$ when $X = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ and $y = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

(b) Is $X^T X$ invertible? Why?